Class X Maths Standard Sample Paper – Set 1

Maximum Marks: 80 **Time Allowed: 3 hours**

General Instructions:

1. This Question Paper has 5 Sections A, B, C, D and E.

2. Section A has 20 MCQs carrying 1 mark each

3. Section B has 5 questions carrying 02 marks each.

4. Section C has 6 questions carrying 03 marks each.

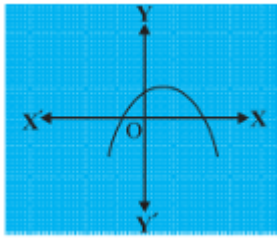
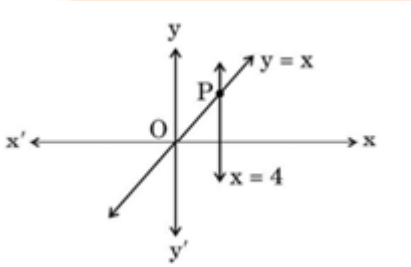
5. Section D has 4 questions carrying 05 marks each. Maximum Marks: 80

6. Section E has 3 case based integrated units of assessment (04 marks each) with sub- parts of the values of 1, 1 and 2 marks each respectively.

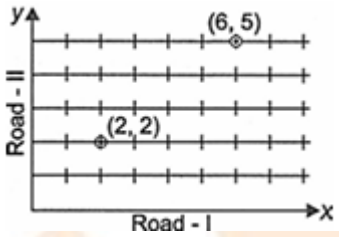
7. All Questions are compulsory. However, an internal choice in 2 Qs of 5 marks, 2 Qs of 3 marks and 2 Questions of 2 marks has been provided. An internal choice has been provided in the 2marks questions of Section E

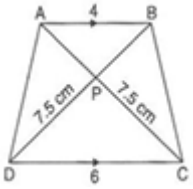
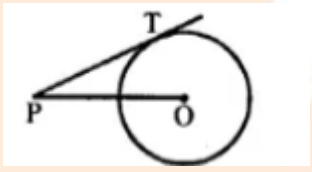
8. Draw neat figures wherever required. Take $\pi = \frac{22}{7}$ wherever required if not stated.

	Section A	
Q. No.	**Questions**	**Marks**
1.	If two positive integers m and n can be expressed as $m = x^2y^5$ and $n = x^3y^2$, where x and y are prime numbers, then HCF $(m, n) =$	1

	a) x^2y^2 b) x^2y^3 c] x^3y^2 d) x^3y^3	
2.	Find the number of zeroes of p(x) in the graph given below.  a) 3 b) 0 c) 2 d) 1	1
3.	The lines represented by the linear equations y = x and x = 4 intersect at P. The coordinates of the point P are:  	1

	a) (4, 4) b) (-4, 4) c) (0, 4) d) (4, 0)	
4.	If y = 1 is a common root of the equations $ay^2+ay+3 = 0$ and $y^2+y+b=0$, then ab equals a) -1 b) -1/p c) 1 d) 1/p	1
5.	if the second term of an AP is 13 and its fifth term is 25, then its 7th term is a) 37 b) 33 c) 38 d) 30	1
6.	A well-planned locality has two straight roads perpendicular to each other. There are 5 lanes parallel to Road - I. Each lane has 8 houses as seen in figure. Chaitanya lives in the 6th house of the 5th lane and	1

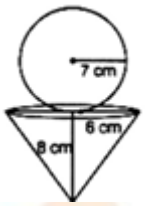
	Hamida lives in the 2nd house of the 2nd lane. What will be the shortest distance between their houses?  a) 12 units b) 5 units c) 6 units d) 10 units	
7.	The vertices of a $\triangle ABC$ are $A(2,1)$, $B(6,-2)$, $C(8,9)$. If AD is the angle bisector of $\angle BAC$, where D meets on BC, then coordinates of D are _____. a) $(5,2)$ b) $\left(\frac{14}{3},\frac{7}{3}\right)$ c) $(4,3)$ d) $\left(\frac{20}{3},\frac{5}{3}\right)$	1
8.	In the given figure, if AB||DC, then AP is equal to	1

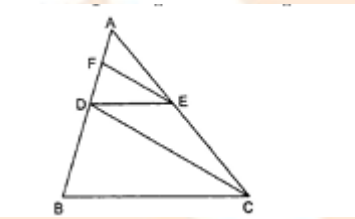
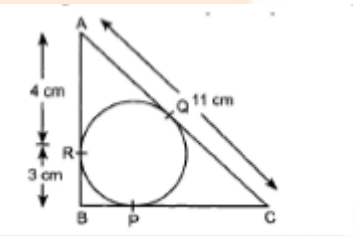
	 a) 5 cm. b) 7 cm. c) 6 cm. d) 5.5 cm.	
9.	In the given figure, point P is 26 cm away from the centre O of a circle and the length PT of the tangent drawn from P to the circle is 24 cm. Then, the radius of the circle is  a) 13 cm b) 10 cm c) 15 cm d) 12 cm	1
10.	AB and CD are two parallel tangents to a circle of radius 5 cm. The distance between the tangents is a) 5cm	1

	b) $\sqrt{50}$cm c) $2\sqrt{5}$cm d) 10cm	
11.	(cosec θ - sin θ) (sec θ – cos θ) (tan θ + cot θ) is equal a) 1 b) 0 c)2 d) -1	1
12.	$\cos^2 30^\circ \cos^2 45^\circ + 4\sec^2 60^\circ + \frac{1}{2}\cos^2 90^\circ - 2\tan^2 60^\circ = ?$ a) $\frac{75}{8}$ b) $\frac{73}{8}$ c) $\frac{83}{8}$ d) $\frac{81}{8}$	1
13.	The angle of elevation of the sun when the shadow of a pole `h' metres high is $\frac{h}{\sqrt{3}}$ metres long is	1

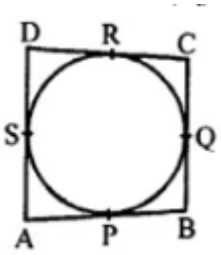
	a) 45° b) 30° c) 60° d) 15°	
14.	The length of the minute hand of a clock is 21 cm. The area swept by the minute hand in 10 minutes is (a) 252 cm^2 (b) 126 cm^2 c) 231 cm^2 d) 210 cm^2	1
15.	Area of a sector of angle p (in degrees) of a circle with radius R is (a) $\frac{p}{360}\times 2\pi R$ b) $\frac{p}{180}\times \pi R^2$ c) $\frac{p}{180}\times 2\pi R$ d) $\frac{p}{720}\times 2\pi R^2$	1
16.	If P(E) = 0.05, what will be the probability of 'not E'?	1

	a) 0.55 b) 0.59 c) 0.95 d) 0.095	
17.	Two dice are rolled together. What is the probability of getting a sum greater than 10? a) $\frac{5}{18}$ b) $\frac{1}{9}$ c) $\frac{1}{6}$ d) $\frac{1}{12}$	1
18.	The arithmetic mean of 1, 2, 3, 4, ..., n is: a) $\frac{n-1}{2}$ b) $\frac{n(n+1)}{2}$ c) $\frac{n}{2}$ d) $\frac{n+1}{2}$	

19.	Assertion (A): A sphere of radius 7 cm is m ounted on the solid cone of radius 6 cm and height 8 cm. the volume of the combined solid is 1737.47 cm^3. [Take $= 3.14$]  Reason (R): Volume of sphere and surface area of cone is given by and $\frac{4}{3}\pi r^3$ and $\frac{1}{3}\pi r^2 h$ respectively. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false d) A is false but R is true	
20.	Assertion (A): Let the positive numbers a, b, c be in A.P., then 1/bc, 1/ac, 1/ab are also in AP Reason (R): If each term of an A.P. is divided by a b c, then the resulting sequence is also in A.P. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true.	

	Section B	
21.	The difference of the square of two numbers is 45. The square of the smaller number is 4 times the larger number. Find the number.	2
22.	In Fig. and DE∥BC CD∥EF. Prove that AD^2 = AB x AF 	2
23.	In figure, ΔABC is circumscribing a circle. Find the length of BC. 	2
24.	If A = B = 60°, verify that sin (A - B) = sin A cos B - cos A sinB OR Prove the trigonometric identity: $tan^2A\ sec^2B - sec^2A\ tan^2B = tan^2A - tan^2B$	2
25.	Four cows are tethered at the four corners of a square field of side 50 m such that each can graze the maximum unshared area. What area will be left ungrazed? [Take = 3.14.]	2

	A sector of a circle of radius 4 cm contains an angle of 30°. Find the area of the sector.	
	Section C	
26.	For a morning walk, three persons steps off together. The measure of their steps is 80,85 and 90 cm respectively.What is the minimum distance each should walk so that all can cover the same distance in complete steps? Which value is preferred in this situation?	3
27.	Find a quadratic polynomial whose sum and product of the zeroes are $\frac{-21}{8}$ and $\frac{5}{16}$ respectively. Also find the zeroes of the polynomial by factorisation.	3
28.	Half of the difference between two numbers is 2. The sum of the greater number and twice the smaller number is 13. Find the numbers. OR Solve algebraically the following pair of linear equations for x and y $31x + 29y = 33$ $29x + 31y = 27$	3
29.	Out of the two concentric circles, the radius of the outer circle is 5 cm and the chord AC of length 8 cm is a tangent to the inner circle. Find the radius of the inner circle. Or	3

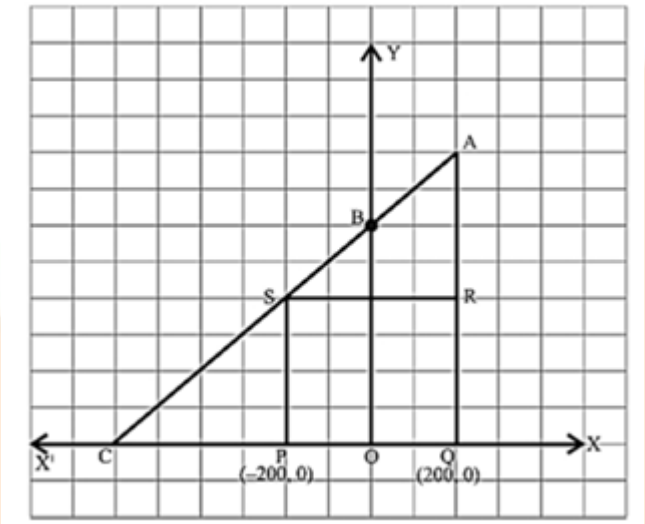
	In the given figure, a quadrilateral ABCD is drawn to circumscribe a circle such that its sides AB, BC, CD and AD touch the circle at P, Q, R and S respectively. If AB = x cm, BC = 7 cm, CR = 3 cm and AS = 5 cm, find x.  	
30.	Prove that: $\frac{1}{\csc A-\cot A}-\frac{1}{\sin A}=\frac{1}{\sin A}-\frac{1}{\csc A+\cot A}$.	3
31.	Five coins were simultaneously tossed 1000 times and at each toss the number of heads were observed. The number of tosses during which 0,1,2,3,4 and 5 heads were obtained are shown in the table below. Find the mean number of heads per toss.	3

No of Heads per toss	No of Tosses
0	38
1	144
2	342
3	287
4	164
5	25
Total	1000

	Section D	
32.	Solve for x $\frac{1}{a+b+x}=\frac{1}{a}+\frac{1}{b}+\frac{1}{x}$ where $a+b+x\neq 0$ and $a,b,x\neq 0$. OR If $x=-4$ is a root of the equation $x^2+2x+4p=0$, find the values of k for which the equation $x^2+px(1+3k)+7(3+2k)=0$ has equal roots.	5
33.	if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio.	5
34.	A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the base of the cylinder or the cone is 24 m. The height of the cylinder is 11 m. If the vertex of the cone is 16 m above the ground, find the area of the canvas required for making the tent. (Use $\pi=\frac{22}{7}$) OR From a cubical piece of wood of side 21 cm, a hemisphere is carved out in such a way that the diameter of the hemisphere is equal to the side of the cubical piece. Find the surface area and volume of the remaining piece.	5
35.	Find the mean of the following frequency distribution:	5

Class	5-15	15-25	25-35	35-45	45-55	55-65
Frequency	6	11	21	23	14	5

	Section E	
36.	Read the text carefully and answer the questions: 5 Kamla and her husband were working in a factory in Seelampur, New Delhi. During the pandemic, they were asked to leave the job. As they have very limited resources to survive in a metro city, they decided to go back to their hometown in Himachal Pradesh. After a few months of struggle, they thought to grow roses in their fields and sell them to local vendors as roses have been always in demand. Their business started growing up and they hired many workers to manage their garden and do packaging of the flowers.  In their garden bed, there are 23 rose plants in the first row, 21 are in the 2nd, 19 in 3rd row and so on. There are 5 plants in the last row. (a) How many rows are there of rose plants? (b) Also, find the total number of rose plants in the garden. OR (c) How many plants are there in 6th row.	4
37.	Read the text carefully and answer the questions: OR Jagdish has a field which is in the shape of a right angled triangle AQC. He wants to leave a space in the form of a square PQRS inside the field for growing	4

	wheat and the remaining for growing vegetables (as shown in the figure). In the field, there is a pole marked as O.  a) Taking O as origin, coordinates of P are (-200, 0) and of Q are (200, 0). PQRS being a square, what are the coordinates of R and S? b) What is the area of square PQRS? Or If S divides CA in the ratio K : 1, what is the value of K, where point A is (200, 800)? c) What is the length of diagonal PR in square PQRS?	
38.	Read the text carefully and answer the questions: An observer on the top of a 40m tall light house (including height of the observer) observes a ship at an angle of depression 30 coming towards the base of the light house along straight line joining the ship and the base of the light house. The angle of depression of ship changes to 45o after 6 seconds.	4



(a) Find the distance of ship from the base of the light house after 6 seconds from the initial position when angle of depression is 45o .

(b) Find the distance between two positions of ship after 6 seconds?

Or

Find the distance of ship from the base of the light house when angle of depression is 30.

c) Find the speed of the ship?

Answer Key

Section A

1. a) x^2y^2

2. (c) 2

3. (a) (4, 4)

4. a) 3

5. (b) 33

6. (b) 5 units

7. d) $\left(\frac{20}{3},\frac{5}{3}\right)$

8. a) 5 cm.

9. (b) 10 cm

10. (d) 10 cm

11. (a) 1

12. (c) $\frac{83}{8}$.

13. c) 60∘

14. (c) 231 cm^2

15. d) $\frac{p}{720}\times 2\pi R^2$

16. (c) 0.95

17. $\frac{1}{12}$

18. $\frac{n+1}{2}$

19. (c) A is true but R is false.

20. (a) Both A and R are true and R is the correct explanation of A

Section B

21.

Let the larger number be x and the smaller number be y . Then,

$x = x^2 - y^2 = 45 \ldots\ldots (i)$

And

$y^2 = 4x \ldots\ldots (ii)$

Substituting $y^2 = 5x$ in (i), we have

$x^2 - 4x = 45$

$x^2 - 4x - 45 = 0$

$x^2 - 9x + 5x - 45 = 0$

$x(x-9) + 5(x-9) = 0$

$(x-0)(x+5)=0$

$\Rightarrow$ **either** $x=0$ or $x+5=0$

$x=9$ or $x=-5$

Smaller number is $\sqrt{36}$ or $\sqrt{-20}$ [From (ii)]

$\Rightarrow$ smaller number is ± 6; $\sqrt{-20}$ is not real.

Larger number =9 and Smaller Number =+_6

22.

In $\triangle ABC$**, it is given that** $DE \| BC \Rightarrow \frac{AB}{AD}=\frac{AC}{AE} \quad \ldots(i)$

In $\triangle ADC$**, it is given that** $FE \| DC \Rightarrow \frac{AD}{AF}=\frac{AC}{AE} \quad \ldots(ii)$

From (i) and (ii), we get $\frac{AB}{AD}=\frac{AD}{AF}$

$\Rightarrow AD^2 = AB \times AF$

23.

AR=4 cm.

Also, AR=AQ $\Rightarrow$AQ=4cm

Now, QC=AC−AQ

=11cm−4cm=7cm….(1)

BP =BR

BP = 3 cm and PC = QC

PC = 7 cm [From (i)]

BC=BP +PC

=3cm+7cm

=10cm

24.

L.H.S. = $\sin(A-B)=\sin(60^\circ-60^\circ)$

$=\sin 0^\circ=0$

R.H.S. = $\sin A\cos B-\cos A\sin B$

$=\sin 60^\circ\cos 60^\circ-\cos 60^\circ\sin 60^\circ$

$=\frac{\sqrt{3}}{2}\times\frac{1}{2}-\frac{1}{2}\times\frac{\sqrt{3}}{2}=0$

Thus, L.H.S. = R.H.S.

OR

$\text{L.H.S.} = \tan 2A \sec 2B - \sec 2A \tan 2B$

$= \tan 2A(1+\tan 2B) - (1+\tan 2A)\tan 2B [\text{using } 1+\tan^2\theta = \sec^2\theta]$

$= \tan^2 A + \tan^2 A \tan^2 B - \tan 2B - \tan^2 A \tan^2 B$

$= \tan^2 A - \tan 2B$

$= \text{R.H.S.}$

Hence proved

25.

Shaded area = area of square - area of sector

$= \left[(50\times 50) - \frac{4\times\pi\times(25)^2\times 90}{360}\right]\text{m}^2$

$= [2500 - 3.14\times 25\times 25\times 0.25]\text{m}^2$

$= [2500 - 1962.5]\text{m}^2$

$= 537.5\text{m}^2$

or

Radius of circle = 4 cm $\theta = 30^\circ$

$\therefore$ Area of sector $= \frac{\theta}{360^\circ}\times\pi r^2$

$= \frac{30^\circ}{360^\circ}\times\pi\times 4\times 4\text{cm}^2$

$= \frac{1}{12} \times \pi \times 16 = \frac{4\pi}{3} \text{cm}^2$.

Section C

26.

Since, the three persons start walking together.

∴ The minimum distance each should walk so that all can cover the same distance in complete steps will be equal to LCM of 80,85 and 90.

Prime factors of 80,85 and are as following

80 =16×5= $2^4 \times 5$

85 =5×17

90 =2×9×5=2× $3^2 \times 5$

So, LCM of 80,85 and 90 = × ×5×17

=16×9×5×17=12240

∴ Each person should walk the minimum distance

=12240 cm =122-meter 40 cm

Value of morning walk: “An early morning walk is a blessing for the whole day.”

27.

We know, quadratic polynomial = x^2 - (Sum of zeroes)$\times x$ + Product of zeroes

Given, Sum of zeroes = $-\frac{21}{8}$ and Product of zeroes = $\frac{5}{16}$

$\therefore$ Quadratic Polynomial $= x^2 + \frac{21}{8}x + \frac{5}{16}$

$= \frac{1}{16}(16x^2 + 42x + 5)$

$\Rightarrow$ Quadratic polynomial is $16x^2 + 42x + 5$

Now, we rewrite the polynomial as $16x^2 + 2x + 40x + 5$

$= 2x(8x+1) + 5(8x+1)$

$= (2x+5)(8x+1)$

Now, for zeroes, $(8x+1)(2x+5) = 0$

$\Rightarrow x = -\frac{1}{8}, -\frac{5}{2}$

28.

Let the greater number be x and the smaller number be y.

It is said that half the difference between the two numbers is 2. So, we can write it as;

$\frac{1}{2}(x-y) = 2$

$\Rightarrow (x-y)=4 \quad \ldots(1)$

It is also said that the sum of the greater number and twice the smaller number is 13. So, we can write it as.

$x+2y=13 \quad \ldots(2)$

Subtracting (2) from (1), we get;

$(x-y)-(x+2y)=4-13$

$\Rightarrow x-y-x-2y=-9$

$\Rightarrow -3y=-9$

$\Rightarrow y=\frac{9}{3}=3$

Putting $y=3$ in (1), we get;

$x-y=4$

$x-3=4$

$\Rightarrow x=7$

Hence, the greater number is 7. The smaller number is 3.

OR

$31x+29y=33 \quad \ldots(1)$

$29x+31y=27 \quad \ldots(2)$

Multiply (1) by 29 and (2) by 31 (Since 29, 31 are primes and LCM is 29×31);

(1) becomes $31x \times 29 + 29x \times 29y = 33 \times 29$ ------- (3)

(2) becomes $29x \times 31 + 31x \times 31y = 27 \times 31$ ------- (4)

Subtracting (3) from (4),

$(312 - 292)y = 27 \times 31 - 33 \times 29 = -120$

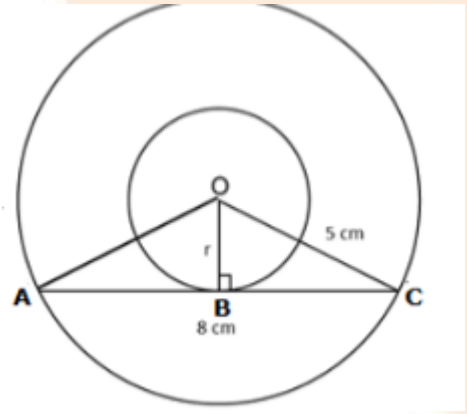
$(31 - 29)(31 + 29)y = -120$

$120y = -120$

$y = -1$

The greater number is $x = 29$ and $y = -1$.

29.



Since AC is a tangent to the inner circle, $\angle OBC = 90^{\circ}$ AC is a chord of the outer circle.

We know that the perpendicular drawn from the centre to a chord of a circle, bisects the chord.

$AC = 2BC$

$8 = 2BC$

$\Rightarrow BC = 4\text{cm}$

In $\triangle OBC$,

By Pythagoras theorem,

$OC^2 = OB^2 + BC^2$

$\Rightarrow 5^2 = r^2 + 4^2$

$\Rightarrow 25 = r^2 + 16$

$\Rightarrow r^2 = 25 - 16$

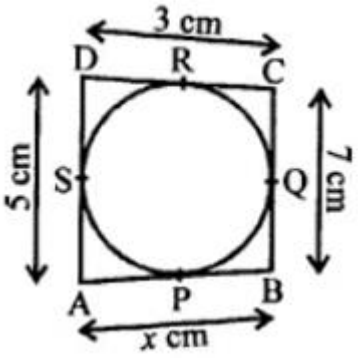
$\Rightarrow r^2 = 9\text{cm}^2$

$\Rightarrow r = 3\text{cm}$

OR

In the given figure, a quadrilateral ABCD is circumscribed a circle touching its sides at P, Q, R, and S respectively.

AB = x cm, BC = 7 cm, CR = 3 cm, and AS = 5 cm.



A circle touches the sides of a quadrilateral ABCD.

AB + CD = BC + AD …(i)

Now, AP and AS are tangents to the circle

AP = AS = 5 cm …(ii)

Similarly, CQ = CR = 3 cm

BP = BQ = x – 5 = 4

BQ = BC – CQ = 7 – 3 = 4 cm

x – 5 = 4

⇒ x = 4 + 5

= 9 cm

30.

We have,

$$\Rightarrow \frac{1}{\csc A-\cot A}-\frac{1}{\sin A}=\frac{1}{\sin A}-\frac{1}{\csc A+\cot A}$$

$$\Rightarrow \frac{1}{\csc A-\cot A}+\frac{1}{\csc A+\cot A}=\frac{1}{\sin A}+\frac{2}{\sin A}$$

$$\Rightarrow \text{LHS}=\frac{\csc A-\cot A}{\csc A+\cot A}+\frac{\csc A+\cot A}{\csc A+\cot A}$$

$$\Rightarrow \frac{\csc^2 A-\cot^2 A}{2\csc A}$$

$$=\frac{\sin A}{1}$$

$$=\frac{2}{\sin A}$$

Hence proved.

31.

No of heads per toss	**No of tosses**	$f_i x_i$
0	38	0
1	144	144
2	342	684
3	287	861
4	164	656

5	25	125
Sum	**1000**	**2470**

Mean number of heads per toss $\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{2470}{1000} = 2.47$

Therefore, Mean = 2.47

Section D

32.

$$\frac{1}{a+b+x} = \frac{1}{a} + \frac{1}{b} + \frac{1}{x}$$

$$\Rightarrow \frac{1}{a+b+x} - \frac{1}{x} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{-(a+b)}{x^2 + (a+b)x} = \frac{b+a}{ab}$$

$$\Rightarrow x^2 + (a+b)x + ab = 0$$

$$\Rightarrow (x+a)(x+b) = 0$$

$$\Rightarrow x = -a, x = -b$$

Hence, $x = -a, -b$.

Let $x = -4$ be the root of the equation $x^2 + 2x + 4p = 0$,

$(-4)^2+(2\times-4)+4p=0$

$or,\ p=-2$

Equation $x^2-2(1+3k)x+7(3+2k)=0$ has equal roots.

$\therefore 4(1+3x)^2-28(3+2k)=0$

or,

$9k^2-8k-20=0$

$or,(9k+10)(k-2)=0$

$\Rightarrow k=\frac{-10}{9},2$

Hence, the value of $k=\frac{-10}{9},2$

33.

Given: ABC is a triangle in which $DE \| BC$.

To prove: $\frac{AD}{BD}=\frac{AE}{CE}$.

Construction: Draw $DN \perp AE$ and $EM \perp AD$, join BE and CD.

Proof:

In $\triangle ADE$,

$$\text{Area of } \triangle ADE = \frac{1}{2} \times AE \times DN \quad \ldots(i)$$

In $\triangle DEC$,

$$\text{Area of } \triangle DCE = \frac{1}{2} \times CE \times DN \quad \ldots(ii)$$

Dividing equation (i) by equation (ii),

$$\Rightarrow \frac{\text{area}(\triangle ADE)}{\text{area}(\triangle DCE)} = \frac{1}{2} \times \frac{AE \times DN}{1} 2 \times CE \times DN$$

$$\Rightarrow \frac{\text{area}(\triangle ADE)}{\text{area}(\triangle DCE)} = \frac{AE}{CE} \quad \ldots(iii)$$

Similarly, in $\triangle ADE$,

$$\text{Area of } \triangle ADE = \frac{1}{2} \times AD \times EM \quad \ldots(iv)$$

In $\triangle DEB$,

$$\text{Area of } \triangle DEB = \frac{1}{2} \times EM \times BD \quad \ldots(v)$$

Dividing equation (iv) by equation (v),

$$\Rightarrow \frac{\text{area}(\triangle ADE)}{\text{area}(\triangle DEB)} = \frac{1}{2} \times \frac{AD \times EM}{1} 2 \times BD \times EM$$

$$\Rightarrow \frac{\text{area}(\triangle ADE)}{\text{area}(\triangle DEB)} = \frac{AD}{BD} \quad \ldots(vi)$$

Since $\triangle DEB$ and $\triangle DEC$ lie on the same base DE and between two parallel lines DE and BC,

$\text{area}(\triangle DEB) = \text{area}(\triangle DEC)$

From equation (iii),

$\Rightarrow \frac{\text{area}(\triangle ADE)}{\text{area}(\triangle DEB)} = \frac{AE}{CE} \quad \ldots(vii)$

Thus, $\frac{AD}{BD} = \frac{AE}{CE}$, as required.

From equation (vi) and equation (vii),

$\frac{AE}{CE} = \frac{AD}{BD}$

$\therefore$ If a line is drawn parallel to one side of a triangle to intersect the other two sides in two points, then the other two

34.

Diameter of cylinder = 24m

$\therefore$ radius of cylinder = radius of cone = 12m

Height of cylinder = 11m

Total height of tent = 16m

$\therefore$ Height of cone = 16 - 11 = 5m

Now, $l^2 = r^2 + h^2$

$\Rightarrow l^2 = 12^2 + 5^2 \Rightarrow l^2 = 144 + 25 = 169$

$\Rightarrow l = \sqrt{169} = 13m$

$\therefore$ Canvas required for tent = curved surface area of cone + curved surface area of cylinder

$= \pi rl + 2\pi rh$

$= \frac{22}{7} \times 12 \times 13 + 2 \times \frac{22}{7} \times 12 \times 11 = \frac{22}{7} \times 12\left[13 + 2 \times 11\right]$

$= \frac{22}{7} \times 12 \times 35 = 22 \times 12 \times 5 = 1320\text{m}^2$

Or

Given side of a cube = 21 cm

Diameter of the hemisphere is equal to the side of the cubical piece (d) = 21 cm

$\Rightarrow$ Radius of the hemisphere = 10.5 cm

Volume of cube = Side^3

$= (21)^3 = 9261\text{cm}^3$

Surface area of cubical piece of wood = $6a^2$

$= 6 \times 21 \times 21\text{cm}^2 = 2646\text{cm}^2$

Volume of the hemisphere = $\frac{2}{3}\pi r^3$

$= \frac{2}{3} \times \frac{22}{7} \times 10.5 \times 10.5 \times 10.5 = 44 \times 0.5 \times 10.5 \times 10.5 \times 10.5 = 2425.5\text{cm}^3$

Surface area of hemisphere = $2\pi r^2$

$= 2 \times \pi \times 10.5 \times 10.5 \text{cm} = 693 \text{cm}^2$

Volume of remaining solid = Volume of cubical piece of wood - Volume of hemisphere

$\Rightarrow$ Volume of the remaining solid $= 9261 - 2425.5 = 6835.5 \text{cm}^3$

Surface area remaining piece of solid = surface area of cubical piece - surface area of hemisphere

$= 6a^2 - \pi r^2 + 2\pi r^2 = (2646 - \pi \times 10.5^2 + 693) \text{cm}^2 = 2992.5 \text{cm}^2$

35.

CI	f_i	x_i	d_i	u_i	$f_i x_i$
5 - 15	6	10	-20	-2	-12
15 - 25	11	20	-10	-1	-11
25 - 35	21	30	0	0	0
35 - 45	23	40	10	1	23
45 - 55	14	50	20	2	28

55 - 65	5	60	30	3	15
Total	**80**				**43**

$$\text{Mean} = A + \frac{\sum u_i}{\sum f_i} \times h$$

$$= 30 + \frac{43}{80} \times 10$$

$$= 35.375$$

Section E

36.

(i) The number of rose plants in the 1st, 2nd, …… are 23, 21, 19, ……5

$a = 23$, d = 21 - 23 = -2, $a_n = 5$

$$\therefore a_n = a + (n-1)d$$

$$\Rightarrow 5 = 23 + (n-1)(-2)$$

$$\Rightarrow 5 = 23 - 2(n-1)$$

$$\Rightarrow 5 = 25 - 2n$$

$$\Rightarrow 2n = 20$$

$$\Rightarrow n = 10$$

(ii) Total number of rose plants in the flower bed,

$S_n = \frac{n}{2}[2a + (n-1)d]$

$S_{10} = \frac{10}{2}[2(23) + (10-1)(-2)]$

$S_{10} = 5[46 - 20 + 2]$

$S_{10} = 5(46 - 18)$

$S_{10} = 5(28)$

$S_{10} = 140$

Or

$S_n = 80$ $S_n = \frac{n}{2}[2a + (n-1)d]$

$\Rightarrow 80 = \frac{n}{2}[2 \times 23 + (n-1) \times (-2)] \Rightarrow 80 = \frac{n}{2}[46 + (n-1)(-2)]$

$\Rightarrow 80 = 23n - n^2 + n \Rightarrow n^2 - 24n + 80 = 0$

$\Rightarrow (n-4)(n-20) = 0 \Rightarrow n = 4$ or $n = 20$

Since $n = 20$ is not possible, we have $n = 4$ For a_{20}:

$a_{20} = 23 + 19 \times (-2) = -15$ The number of plants cannot be negative. Thus, $n = 4$.

(iii) $a_n = a + (n-1)d$

For a_6: $a_6 = 23 + 5 \times (-2)$ $a_6 = 13$

37.

(i) Since, PQRS is a square

$\Rightarrow PQ = QR = RS = PS$ Length of $PQ = 200 - (-200) = 400$

$\Rightarrow$ The coordinates of $R = (200, 400)$ and coordinates of $S = (-200, 400)$

(ii) Area of square PQRS = $(\text{side})^2$

$= (400)^2 = 160,000$sq. units

Since, point S divides CA in the ratio K : 1,

$$\frac{Kx+1}{K+1}, \frac{Ky+1}{K+1} = (-200, 400) \Rightarrow \frac{K(200)+(-600)}{K+1}, \frac{K(800)+0}{K+1} = (-200, 400)$$

$$\Rightarrow \frac{200K-600}{K+1} = -200, \frac{800K+400}{K+1} = 400 \Rightarrow 800K + 400 = 400(K+1)$$

$$\Rightarrow 400K = 400 \Rightarrow K = 1$$

(iii) By Pythagoras theorem

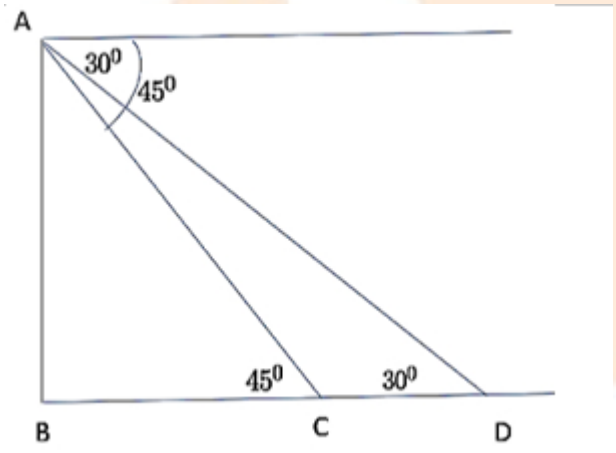
$(PR)^2 = (PQ)^2 + (QR)^2 = 400^2 + 400^2$

$= 160,000 + 160,000 = 320,000$

$\Rightarrow PR = \sqrt{320,000} = 400 \times \sqrt{2}$ units

38.

(i)



The distance of the ship from the base of the lighthouse after 6 seconds from the initial position when the angle of depression is 45°.

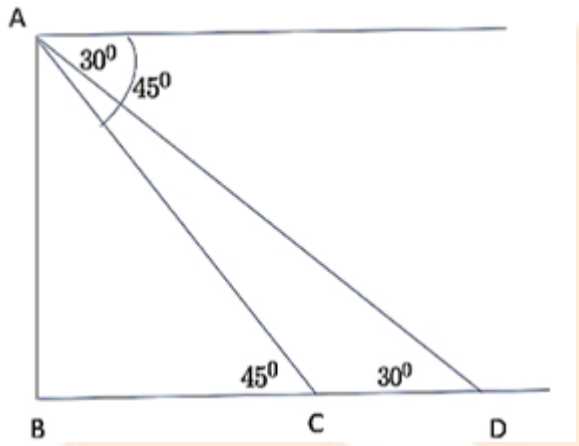
In $\triangle ABC$,

$\tan 45^{\circ} = \frac{AB}{BC}$

$\Rightarrow 1 = \frac{40}{BC}$

$\Rightarrow BC = 40\text{m}$

(ii)



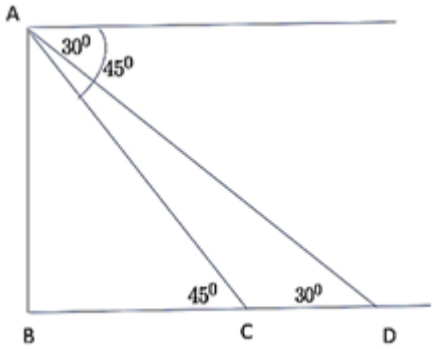
The distance between two positions of the ship after 6 seconds.

$CD = BD - BC$

$\Rightarrow CD = 40 \times \sqrt{3} - 40 = 40\left(\sqrt{3} - 1\right)$

$\Rightarrow CD = 29.28\text{m}$

Or



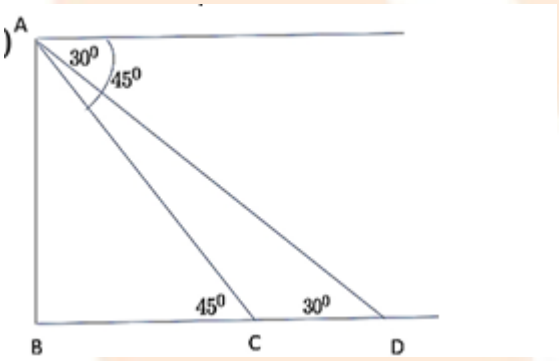
The distance of the ship from the base of the lighthouse when the angle of depression is 30°.

In $\triangle ABD$,

$$\tan 30^{\circ} = \frac{AB}{BD} \Rightarrow \frac{1}{\sqrt{3}} = \frac{40}{BD}$$

$$\Rightarrow BD = 40 \times \sqrt{3}\text{m}$$

(iii)



$$\text{Speed of ship} = \frac{\text{Distance}}{\text{Time}}$$

$$\text{Speed of ship} = \frac{29.28}{6} = 4.88\text{m/sec}$$